MM-COMPLIANCE: A Multilingual Multimodal Dataset for Sustainability Compliance Auditing

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Abstract

Detecting whether corporate sustainability disclosures truly satisfy reporting standards is essential for greenwashing-aware ESG auditing, yet existing resources rarely evaluate whether disclosures sufficiently satisfy framework metrics. We introduce MM-COMPLIANCE, a multilingual multimodal dataset aligned with SASB metrics across six languages (English, French, Japanese, Chinese, Thai, and Korean), with page-level evidence and a key *completeness* annotation that separates fully compliant disclosures from partial mentions. We define two tasks reflecting practical auditing workflows: Page-level Compliance Judgment and Metric-to-Page Evidence Retrieval. Our evaluated baselines primarily comprising open-source architectures, supplemented by a commercial VLM as an upper-bound reference, show moderate performance for page understanding and evidence localization, but consistently low performance on partial compliance verification and large variation across languages and industries. MM-COMPLIANCE provides a benchmark for developing framework-aware models to support rigorous ESG compliance auditing and long-term commitment tracing.[1]

1 Introduction

Companies worldwide are under increasing pressure to back up their lofty sustainability pledges with concrete results. Yet a troubling gap often exists between ESG (Environmental, Social, and Governance) commitments and actual performance. Studies have found that nearly 40% of firms setting 2020 climate targets either failed to meet their goals or quietly stopped reporting on them – and faced virtually no consequences (Jiang et al., 2025). In the investment sphere, over half of ESG-focused funds have been accused of exaggerating their claims, with 70% falling short of their own ESG targets (Montgomery et al., 2024). This lack of accountability enables greenwashing, where organizations reap reputational benefits from announcing goals but suffer no penalty when progress stalls. The result is an erosion of stakeholder trust and a dilution of genuine sustainability efforts.

To combat greenwashing, regulators and investors are increasingly demanding transparency through standardized reporting. One prominent framework is the Sustainability Accounting Standards Board (SASB), which defines industry-specific metrics for disclosing financially material ESG information. In theory, if companies rigorously report these metrics, stakeholders can trace commitments to measurable outcomes – a process we might call commitment tracing or ESG transparency. In practice, however, such reporting is inconsistent. Companies often provide only partial data or vague statements instead of the precise figures required by standards, making it difficult to verify whether a commitment (e.g. reducing carbon emissions or improving workplace safety) has been fulfilled. Without tools to systematically track these disclosures, even well-intentioned frameworks like SASB cannot fully shine light on corporate sustainability performance.

A major obstacle is the multimodal and multilingual nature of modern sustainability reporting. Unlike traditional financial statements, ESG reports are lengthy, glossy publications filled with dense text, multi-column layouts, intricate tables, and colorful charts (Seki et al., 2025). Crucial details might appear in a footnote, an infographic, or a complex KPI table rather than in plain sentences. This makes it challenging for text-only NLP systems to find and interpret the relevant information. Moreover, global companies publish these reports in many languages, and direct translations

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[1]Project Link: https://haalab.github.io/projects/mm-compliance/

Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing, pages 18675–18690
October 24-29, 2026 ©2026 Association for Computational Linguistics

of SASB metrics can vary, complicating retrieval. A recent study by Gupta et al. (2025) demonstrated that even state-of-the-art vision-language models struggle with extracting information from such visually rich reports, underscoring the need for better multimodal approaches.

In this paper, we address these challenges by presenting a new multilingual, multimodal dataset for sustainability compliance auditing. We introduce MM-COMPLIANCE, a multimodal ESG dataset spanning six languages (English, French, Japanese, Chinese, Thai, and Korean) across diverse corporate sectors. Crucially, we annotate the exact locations of SASB-defined metrics and evaluate their disclosure completeness, distinguishing genuine framework compliance from partial or superficial mentions.

Our contributions establish a framework for automated ESG commitment tracking and transparency analysis. First, we introduce the MM-COMPLIANCE dataset, the first to our knowledge that aligns SASB's investor-centric metrics with actual disclosure pages in multiple languages. Second, we propose benchmark tasks – including evidence retrieval and compliance classification – that test a model's ability to navigate complex report layouts and verify disclosure completeness. Third, we provide baseline results using vision-language models, illustrating both the potential of multimodal techniques and the remaining gaps (e.g. distinguishing partially fulfilled metrics). By rigorously evaluating whether companies report what they promised to, our work aims to support auditors, analysts, and stakeholders in tracing ESG commitments over time. This not only aids in identifying greenwashing early on, but also encourages better reporting practices – pushing sustainability discourse from mere promises to proven progress.

To ensure reproducibility, the dataset and system implementation code for both benchmark tasks are publicly available at `https://haalab.github.io/projects/mm-compliance/`.

2 Related Work

2.1 Standard-Aligned ESG Disclosures and Benchmarks

Recent years have seen a surge in computational resources for corporate ESG disclosures. Early benchmarks primarily focused on text-only KPI extraction (Tseng et al., 2023; Chen et al., 2023a,b, 2024; Zou et al., 2025; Bora et al., 2025) or document-level retrieval over financial texts (Schimanski et al., 2024; Choi et al., 2025). To map corporate text to sustainability frameworks, studies have aligned disclosures with the Global Reporting Initiative (Ahmed et al., 2025) or the UN Sustainable Development Goals (Wan, 2025). Modern expert-level benchmarks like ESGenius (He et al., 2025) offer multi-framework QA across GRI, SASB, and TCFD. Simultaneously, multimodal benchmarks such as MMESGBench (Zhang et al., 2025) have advanced multi-image reasoning over visually rich report layouts. However, existing datasets remain largely English-centric or text-dominant, failing to address industry-specific, investor-oriented standards across diverse languages. MM-COMPLIANCE bridges this gap by establishing a multilingual, multimodal benchmark explicitly aligned with sector-specific SASB metrics.

2.2 Verification of Disclosure Completeness and Greenwashing

To combat corporate greenwashing, an emerging line of research targets the verification of sustainability commitments. Seki et al. (2025) introduced ML-PROMISE to track corporate promises over time, while the A3CG benchmark (Ong et al., 2025) utilizes aspect-action analysis to differentiate vague corporate plans from executed actions. Despite these advances, a recent meta-analysis by Calamai et al. (2025) revealed that 96% of existing climate NLP datasets suffer from severe annotation noise, enabling models to exploit surface-level lexical shortcuts (i.e., keyword matching) rather than demonstrating genuine framework comprehension. MM-COMPLIANCE tackles this shortcut challenge through a two-fold design: (1) a "proximity hard negative sampling" strategy that forces models to discern subtle metric boundaries on adjacent report pages, and (2) a three-class evaluation scheme (*yes*, *yes but not complete*, *no*) that strictly decouples superficial mentions from standard-compliant, verifiable quantitative disclosures.

2.3 Multi-Page Visually-Rich Document Understanding and Visual RAG

Because corporate ESG disclosures rely heavily on rich visual structures—such as dense multi-column layouts, financial tables, and infographics—text-only extraction inevitably incurs substantial information loss (Seki et al., 2025). While Visually-Rich Document Understanding (VRDU) has ad-

vanced rapidly, practical compliance auditing faces the challenge of Multi-Page Visually-Rich Document Understanding (MP-VRDU), where sparse evidence must be localized across long publications. To circumvent brittle OCR pipelines, recent vision-language retrieval frameworks directly embed document page images. These include single-vector VLM architectures like VisRAG (Yu et al., 2024) as well as image-native late-interaction models such as ColPali (Faysse et al., 2025) and ColQwen2.5. MM-COMPLIANCE bridges the gap between state-of-the-art MP-VRDU architectures and practical auditing workflows through its "Metric-to-Page Evidence Retrieval" task, establishing a rigorous benchmark for VLMs and image-native retrievers in challenging, multilingual settings.

3 Dataset Construction

This section describes the construction of MM-COMPLIANCE, including data sources, metric selection, annotation design, and dataset statistics. The dataset is designed to support compliance-oriented analysis of ESG disclosures, with a focus on identifying incomplete or superficial reporting that may hinder long-term commitment tracing.

3.1 Source Documents

We construct MM-COMPLIANCE from publicly available ESG and sustainability reports released by corporations across multiple industries. All documents are collected in PDF format to preserve their original layout, visual structure, and multimodal content. Unlike traditional financial statements, ESG reports frequently contain tables, charts, infographics, and complex page layouts, which are critical for understanding sustainability disclosures but are often lost in text-only representations.

To ensure a representative and globally balanced corpus, we systematically collect ESG and sustainability reports from six targeted countries/regions: the United Kingdom, France, Taiwan, Japan, Thailand, and South Korea. For each country, we select three major industrial sectors, and within each sector, we choose three distinct public corporations representing the highest (top), median (middle), and smallest (bottom) market capitalization layers. This experimental design yields a minimum of three firms per sector and a total baseline of at least nine unique firms per country, ensuring that our data alignment captures organizational variability across geopolitical and corporate-scale dimensions[2]. This formalized selection methodology aligns strictly with established standards for multilingual corporate disclosure benchmarking.

3.2 SASB Metric Selection

To enable standardized and investor-oriented compliance analysis, we align disclosures with the Sustainability Accounting Standards Board (SASB) framework. SASB defines industry-specific sustainability topics and metrics that emphasize financial materiality and quantitative rigor. For each company, we identify the relevant SASB standards based on its industry and use the corresponding Topic–Metric pairs as queries for annotation.

Both *Topic* and *Metric* are represented as natural language descriptions rather than symbolic codes. This design choice reflects real-world auditing scenarios, where analysts and automated systems must interpret metric definitions expressed in text rather than relying on structured identifiers.

For each targeted corporation, we map the corresponding SASB metrics by first determining the specific industry standards applicable to its primary commercial operations. A potential point of ambiguity in multilingual auditing is how compliance requirements are matched across reports published in different localized languages. To address this framework-to-practice mapping gap, we leverage the fact that the SASB framework taxonomy provides cross-lingually synchronized criteria and standardized natural language descriptions.

Rather than relying on brittle, symbolic alphanumeric codes or automated translation heuristics, we utilize these harmonized cross-lingual framework descriptions directly as text queries for the disclosure alignment process. Because the underlying semantic definition of each target metric is kept strictly identical across language boundaries, annotators are able to evaluate localized source texts (e.g., Japanese or Thai reports) against the same conceptual requirement checklist. This design ensures robust metric alignment and eliminates interpretation bias when evaluating compliance verification models in cross-lingual retrieval scenarios.

3.3 Annotation Scheme and Completeness Labels

The core annotation objective of MM-COMPLIANCE is to assess whether a given report page provides sufficient information to

[2]For some countries and languages, the number of target industrial sectors was adjusted upward from three.

satisfy a specific SASB metric. For each (Topic, Metric, Page) triple, annotators determine whether the disclosure on the page is *complete*, meaning that it fulfills the metric's requirements at a semantic level, including necessary quantitative values, scope definitions, or procedural details.

If a page only partially mentions the metric—such as providing qualitative statements without required numerical data, or reporting limited scope information—it is labeled as *incomplete*. Pages that do not contain relevant information for the metric are excluded from completeness annotation and are used as negative instances in downstream tasks.

This completeness-oriented labeling scheme directly supports greenwashing-aware analysis. By distinguishing full compliance from partial or superficial disclosure, the dataset enables systematic identification of cases where companies appear to address a sustainability topic but fail to provide verifiable, standard-compliant evidence.

3.4 Dataset Statistics

Table 1 summarizes the statistics of MM-COMPLIANCE across languages. The dataset contains a total of 7,174 PDF pages and 366 metric–language combinations. The distribution of complete versus incomplete disclosures varies substantially across languages, reflecting differences in reporting practices, regulatory environments, and document structures.

Notably, Japanese and Korean reports exhibit a higher proportion of incomplete disclosures, suggesting greater challenges for compliance verification in these contexts. In contrast, Chinese-language reports from Taiwanese firms and French reports show higher completeness rates, potentially due to more standardized presentation of sustainability metrics. The inclusion of Thai reports significantly expands linguistic coverage and introduces additional variability in layout and disclosure styles, further motivating the need for multilingual and multimodal analysis.

The relatively small number of annotated instances in the Korean (177) and Thai (123) subsets reflects structural boundaries of SASB-aligned disclosure in these regions, not a collection limitation. In 2022, only 104 Korean companies referenced SASB in their voluntarily disclosed sustainability reports. ESG disclosure mandates in Korea have been officially postponed to post-2026. Thailand has similarly not adopted SASB as a mandatory framework.

Language	#PDF Pages	Metric		Completeness	
		Types (M)	Total (N)	%True	%False
English	682	70	538	74.9	25.1
French	435	61	416	78.8	21.2
Japanese	1,394	50	246	51.6	48.4
Chinese	1,799	99	301	85.4	14.6
Thai	2,017	61	123	78.0	22.0
Korean	847	25	177	59.9	40.1

Table 1: Dataset statistics of MM-COMPLIANCE by language, including the number of PDF pages, distinct SASB metrics, annotated instances, and the proportion of complete and incomplete disclosures.

Crucially, this programmatic sparsity is driven by institutional alignment and structural disclosure factors rather than a deficiency in overall document volumes; while these subsets encompass a substantial number of total source pages (847 for Korean and 2,017 for Thai), a smaller percentage of pages can be confidently aligned with explicit SASB metric definitions. This lower effective density occurs because corporate disclosures in these markets are frequently organized around alternative international frameworks (e.g., GRI) or rely heavily on localized terminology and regional visual elements that do not map directly onto the specific phrasing of the SASB taxonomy. This structural misalignment between domestic disclosure customs and investor-oriented global standards is consistent with the report in other non-Western jurisdictions (e.g., Japan (PwC Japan, 2024)).

Evaluation Benchmark Design: Given these diverse data characteristics and real-world compliance nuances, MM-COMPLIANCE is explicitly formulated as an evaluation benchmark to track framework-aware compliance and reasoning in a zero-shot or few-shot setting, rather than a dataset intended for training models from scratch. Rather than fitting models on domain-specific splits, the benchmark leverages this rich dataset to rigorously measure how general-purpose large vision-language models (VLMs) handle complex, multi-page corporate auditing workflows out of the box.

4 Tasks: From Page Understanding to Report Auditing

The MM-COMPLIANCE dataset is designed to support two complementary evaluation settings that reflect realistic sustainability compliance auditing workflows. In practical auditing workflows, auditors first locate candidate evidence pages and then verify whether the retrieved disclosures fully satisfy the corresponding requirements. For controlled

evaluation, however, we present the page-level verification task first.

4.1 Page-level Compliance Judgment

The first task focuses on page-level semantic understanding and serves as a controlled test of reading comprehension over ESG report pages. Given a single page and a SASB metric definition, the goal is to determine whether the required information is disclosed on that page and whether the disclosure satisfies the metric in full. This setting isolates the verification step of compliance auditing and allows us to analyze multilingual and multimodal understanding without the confounding difficulty of document-level search.

Each instance consists of a SASB Topic–Metric pair and a report page, represented as OCR-extracted text and/or a rendered page image. The model predicts one of three labels: *yes*, *yes but not complete*, or *no*. The label *yes* indicates that the page fully satisfies the SASB metric requirements. The label *yes but not complete* indicates that the metric is mentioned but key required elements are missing, such as quantitative values, scope definitions, or procedural details. The label *no* indicates that the page does not disclose the metric.

To construct challenging negative instances for each target metric, we implement a localized sampling strategy: for every report page that contains a validated complete or partial disclosure, we explicitly sample a nearby page from the same document that does not contain the target SASB metric. This choice of "nearness" is a deliberate design decision intended to introduce *Hard Negatives*, a standard and rigorous practice in representation learning and fine-grained information retrieval. While distant pages often result in "Easy Negatives" due to significant context shifts (e.g., separating Social and Governance sections) , selecting near pages forces the model to discern fine-grained SASB metric nuances within the same section.

4.2 Metric-to-Page Evidence Retrieval

The second task models a more realistic post-publication auditing scenario. After an ESG report is released, auditors typically do not know where a specific metric is disclosed and must search through the entire document to locate the relevant evidence. This task evaluates whether a system can identify the correct disclosure page(s) given only a metric description and a long ESG report.

Given a SASB metric description as a query and a target ESG report consisting of multiple pages, the system ranks all pages in the report by relevance. Each query is associated with a set of ground-truth evidence pages annotated in MMCOMPLIANCE. Retrieval quality is evaluated at multiple cutoffs to measure whether at least one relevant page is retrieved and how highly it is ranked.

To avoid conflating page localization with report identification, retrieval is restricted to pages within the same report. This document-boundary setting mirrors practical auditing workflows, where the report under inspection is known but the evidence locations are not.

Taken together, the two tasks capture complementary aspects of sustainability compliance auditing. Page-level Compliance Judgment evaluates whether a system can verify disclosure completeness once a candidate page is given. Metric-to-Page Evidence Retrieval evaluates whether a system can locate such candidate pages within a long, multimodal document. By combining these tasks, MMCOMPLIANCE supports systematic evaluation of disclosure verification, evidence localization, and long-term ESG commitment tracing.

5 Experimental Setup

To ensure data privacy and reproducibility in enterprise auditing, we prioritize open-source, locally deployable baselines over commercial APIs. We adopt two recent state-of-the-arts: (1) EVisRAG (an updated VisRAG) and (2) ColQwen2.5 (an image-native extension of ColPali), both representing highly competitive standards for multimodal processing.[3]

5.1 Task 1: Page-level Compliance Judgment

The goal of Page-level Compliance Judgment is to assess whether a model can correctly interpret the content of a single report page with respect to a given SASB Topic–Metric pair. This task primarily evaluates document reading and local reasoning ability, rather than evidence search.

Each instance consists of a SASB metric description and a single ESG report page. The model predicts one of three labels: *yes*, *yes but not complete*, or *no*. Evaluation is conducted using overall classification accuracy and F1 score.

[3]Additionally, open-weight VLMs are evaluated in Appendix B for architecture generalizability, while a proprietary model serves strictly as an upper-bound reference to contextualize the performance ceiling of locally deployable models.

We evaluate EVisRAG under four settings with varying modalities and retrieval strategies: **Text-only (w/o RAG)** uses OCR-extracted text from the target page as input to form a minimal reading comprehension baseline. **Image-only (w/o RAG)** utilizes the rendered page image directly, evaluating compliance inference from layouts, tables, and visual structures without OCR. For retrieval-augmented setups, **Text-only (w/ RAG)** and **Image-based (w/ RAG)**, we select the most similar examples from a knowledge base using text similarity (on OCR) or visual similarity, injecting them as few-shot in-context demonstrations before prediction.

5.2 Task 2: Metric-to-Page Evidence Retrieval

Metric-to-Page Evidence Retrieval models a realistic auditing scenario in which the evidence pages for a given SASB metric are unknown in advance. Given a metric description and a multi-page ESG report, the system ranks all pages in the report by relevance and is evaluated by standard retrieval metrics at multiple cutoffs.

We adopt ColQwen2.5 as our baseline, an image-native document retriever that embeds metric queries and PDF page images into multi-vector representations and computes relevance scores via late-interaction matching. We compared ColQwen2.5 and ColPali on the English subset; ColQwen2.5 showed substantially higher retrieval quality and was therefore used across all languages. For each report, we render all PDF pages into images and index them at the page level. At inference time, the metric description is used as the query and the retriever ranks pages within the same report.

To isolate page localization from report identification, retrieval is restricted to the pages of the associated target report (document-boundary retrieval). We report retrieval quality at cutoffs $K \in \{1, 5, 10, 20\}$ using coverage-sensitive metrics, including F1@K and Hit@K.

5.3 Data Splitting, Evaluation, and Hyperparameters

Data Splitting Strategy: Aligning with the benchmark-centric design established in Section 3.4, we omit standard train/val/test splits for supervised model training and instead evaluate zero-shot and few-shot capabilities out of the box. Specifically, for Task 1, to prevent information leakage, all experiments adopt a document-level split stratified by company and language, ensuring that pages from the same report never appear in both the test evaluation set and the retrieval knowledge base. We followed the off-the-shelf configuration released by EVisRAG without adjusting the hyperparameters.

For Task 2, we evaluate retrieval performance using standard ranking metrics, including Hit@K and F1@K, following prior work on document retrieval. We use the default ColQwen2.5 configuration with dynamic resolution (max 768 patches per page) and publicly released weights. All experiments are run on NVIDIA A100 80GB with batched page encoding (average inference time: 11.70s per query).

6 Results

This section reports baseline performance on the two tasks supported by MM-COMPLIANCE. We emphasize cross-lingual variation and the role of modality and retrieval, since both are central to practical ESG auditing and commitment tracing.

6.1 Page-level Compliance Judgment

Table 2 summarizes accuracy for Page-level Compliance Judgment across languages and method settings. Overall, direct page understanding without retrieval provides the strongest and most stable performance. Averaged across languages, the text-only setting reaches 0.58 accuracy, matching the image-only setting (0.58), while retrieval-augmented settings underperform (0.52 for Text+RAG; 0.51 for Image+RAG).

We observe clear language-dependent modality effects. For Chinese-language (Taiwan) and Korean subsets, image-only inference achieves the best performance (0.73 and 0.59, respectively), consistent with the prevalence of table- and figure-heavy disclosures where OCR text can lose layout cues. In contrast, English, French, Japanese, and Thai favor text-based inference, suggesting that metric-related evidence in these subsets is more frequently expressed in running text.

Across languages, retrieval augmentation does not yield consistent gains when the target page is already given. This suggests that Page-level Compliance Judgment primarily tests local reading and completeness verification, where noisy or weakly matched few-shot exemplars can introduce distraction rather than additional evidence.

6.2 Metric-to-Page Evidence Retrieval

Table 3 reports cross-lingual performance for Metric-to-Page Evidence Retrieval, where the sys-

Language	Text	Text w/ RAG	Image	Image w/ RAG
English	0.51	0.50	0.50	0.43
French	0.57	0.51	0.55	0.49
Japanese	0.58	0.55	0.55	0.54
Chinese	0.67	0.55	**0.73**	0.57
Thai	0.63	0.54	0.54	0.54
Korean	0.52	0.44	**0.59**	0.46
Average	**0.58**	0.52	**0.58**	0.51

Table 2: Page-level Compliance Judgment accuracy across languages.

tem must identify evidence pages within a multi-page ESG report given only a SASB metric query. This task is substantially more challenging than page-level judgment, especially at strict cutoffs. On average, the retriever achieves F1@1 of 0.319 and Hit@5 of 0.719, indicating that evidence pages are often retrievable but not reliably ranked at the very top. At larger cutoffs, performance improves markedly (Hit@20 = 0.882), reflecting that relevant pages are frequently present within the top-20 list even when early ranking remains imperfect.

Performance varies substantially by language. French exhibits the strongest retrieval effectiveness at moderate cutoffs (F1@5 = 0.430; Hit@20 = 0.967), which is partly associated with a higher average number of relevant pages per query (Avg.R = 3.14). Chinese achieves the highest evidence coverage at large cutoffs (Hit@20 = 0.991) despite more modest early ranking (F1@1 = 0.375), suggesting that relevant pages are typically retrieved but not always prioritized. Thai is the most challenging subset overall (F1@1 = 0.222; Hit@20 = 0.749), consistent with Avg.R being close to 1, which limits the benefit of multi-page retrieval and amplifies the effect of ranking errors.

Taken together, these results highlight a practical distinction between the two tasks. When the candidate page is already known, strong multilingual models can often perform reasonable local compliance judgment without retrieval. However, in realistic auditing scenarios, evidence localization remains a key bottleneck, and performance depends strongly on language-specific disclosure conventions and document layouts.

To ensure that our findings are not inherent to a specific architecture, we conducted additional evaluations using Qwen3-VL-8B-Instruct, Llama 3.2 Vision, and Gemini 3.5 Flash (serving as an upper-bound performance baseline). The results across both tasks are consistent with our primary findings (see Appendix B for details), confirming

Language	Avg.R	F1@1	F1@5	F1@20	Hit@5	Hit@20
English	1.98	0.421	0.313	0.146	0.757	0.850
French	3.14	0.360	0.430	0.213	0.905	0.967
Chinese	1.50	0.375	0.306	0.123	0.810	0.991
Japanese	1.67	0.252	0.242	0.119	0.651	0.878
Korean	2.66	0.286	0.318	0.161	0.729	0.857
Thai	1.00	0.222	0.154	0.071	0.461	0.749
Average	1.99	0.319	0.294	0.139	0.719	0.882

Table 3: Metric-to-Page Evidence Retrieval performance across languages. We report F1@K and Hit@K at representative cutoffs.

Modality	Retrieval	F1 (yes)	F1 (yes but not complete)	F1 (no)
Text	w/o RAG	0.51	0.06	0.69
	w/ RAG	0.37	0.01	0.64
Image	w/o RAG	0.52	0.05	0.69
	w/ RAG	0.40	0.00	0.62

Table 4: Macro-averaged class-wise F1 scores for Page-level Compliance Judgment.

that the observed challenges in partial compliance are generalized across different VLMs.

7 Discussion

In this section, we organize the discussion into several aspects that correspond to major bottlenecks revealed by MM-COMPLIANCE.

7.1 Fine-grained Completeness Verification Remains Unsolved

The class-wise analysis (Table 4) reveals a systematic weakness that is not apparent from accuracy alone. While models achieve reasonable F1 scores for the complete disclosure class (*yes*) and the absence class (*no*), performance on the partial disclosure class (*yes but not complete*) is consistently close to zero across all modalities and retrieval settings. This failure mode is especially important for greenwashing-aware analysis, because partial disclosures correspond to cases where a report mentions a sustainability topic but omits the elements required for verification, such as quantitative values, scope definitions, or methodology.

These results indicate that completeness verification requires requirement-aware reasoning rather than topic matching. Many partial disclosures share keywords with metric descriptions, but do not satisfy mandatory components. A model must identify what is missing and judge sufficiency under the standard, which is closer to checklist-style validation than conventional document classification. The strong performance on the *no* class combined with near-zero performance on the partial class suggests a conservative strategy: when expected

Gold	Image-based w/o RAG			Text-based w/o RAG		
	yes	yes but not comp.	no	yes	yes but not comp.	no
yes	187	6	210	174	8	221
yes but not comp.	53	5	96	47	7	100
no	45	3	376	36	1	387

Table 5: Confusion matrix for Page-level Compliance Judgment across baseline configurations.

	English			French			Chinese					Japanese			Korean			Thai		
Metric	Ene.	Fin.	Lux.	Ene.	Fin.	Lux.	Bio.	Ene.	Fin.	Sem.	Tec.	Aut.	Ene.	Tra.	Aut.	Har.	Sem.	Ban.	Com.	Ene.
F1@1	**0.535**	0.417	0.293	**0.430**	0.238	0.405	0.338	0.299	0.386	**0.516**	0.327	**0.437**	0.087	0.205	**0.391**	0.116	0.337	0.261	0.132	**0.273**
F1@5	0.330	**0.407**	0.190	0.382	0.319	**0.584**	0.253	0.247	0.352	**0.364**	0.311	**0.301**	0.245	0.174	0.344	0.248	**0.355**	0.159	0.119	**0.182**
F1@10	0.213	**0.323**	0.155	0.259	0.268	**0.425**	0.149	0.200	0.210	**0.233**	0.194	**0.226**	0.200	0.137	0.247	0.203	**0.270**	0.099	0.089	**0.132**
F1@20	0.126	**0.208**	0.100	0.162	0.231	**0.245**	0.101	0.127	0.127	**0.137**	0.124	0.133	**0.134**	0.091	0.150	0.152	**0.182**	0.058	0.074	**0.082**

Table 6: Industry-level Metric-to-Page Evidence Retrieval performance across languages. F1@K scores are reported at multiple cutoffs. For each language and cutoff, the best-performing industry is highlighted in bold. Industry abbreviations: Ene.=Energy, Fin.=Finance, Lux.=Luxury, Bio.=Biomedical, Sem.=Semiconductor, Tec.=Technology, Aut.=Automobile, Tra.=Trading, Har.=Hardware, Ban.=Banking, Com.=Commerce.

numerical formats or explicit KPI structures are not detected, the model defaults to *no* rather than predicting *yes but not complete*.

Table 5 confirms this pattern: most partial disclosures (*yes but not complete*) are systematically misclassified as non-disclosures (*no*), whereas false positives (*yes*) remain rare. This bias is heavily driven by cross-lingual terminology gaps. For example, in Japanese corporate reports (e.g., *Marubeni*), native terms for turnover rate obscure alignment with English SASB metric definitions, triggering false negatives.

7.2 Insights from the NTCIR-19 RegCom Task

The use of MM-COMPLIANCE in the NTCIR-19 RegCom task provides complementary evidence for the failure modes observed above (Chen et al., 2026). Participants explored a range of retrieval and verification strategies, including lexical and semantic retrieval, table-of-contents-guided routing, text and page-image embeddings, multimodal evidence fusion, VLM-based reranking, and retrieval-augmented few-shot prompting. Their results show that incorporating visual evidence and model-based reranking generally improves evidence localization, while metric-aware demonstrations can improve page-level verification, although the gains are not consistent across all settings. Across systems, the *yes but not complete* class remains the most difficult to identify, and end-to-end performance is substantially affected by the propagation of retrieval errors into the final compliance decision. These findings further support our conclusion that requirement-aware completeness reasoning, rather than broad topical matching, is the central challenge, while motivating the industry-level analysis of retrieval performance that follows.

7.3 Industry Effects in Metric-to-Page Retrieval

As reported in Table 6, industry-specific differences are prominent and introduce substantial within-language variability. Crucially, the sharp performance gains from $K{=}1$ to $K{=}10$ suggest that while precise early ranking remains challenging, the retriever is effective as a shortlisting mechanism for practical auditing. This demonstrates that systems can surface relevant pages within a small candidate set, though they remain unreliable for selecting the single best page without verification.

These performance disparities reflect how different sectors format disclosures. Structured, KPI-heavy sectors such as English Energy ($F1@1 = 0.535$) and Chinese Semiconductor ($F1@1 = 0.516$) achieve strong early localization due to standardized tables. Conversely, sectors relying on narrative prose or visual infographics present severe ranking challenges. For instance, Japanese Energy exhibits a low $F1@1$ (0.087) despite a higher $F1@5$ (0.245), indicating top-rank bottlenecks from localized terminology. Similarly, Thai Commerce underperforms Thai Energy because key metrics are scattered across infographics. Developing industry-aware retrievers that leverage visual layout cues is therefore essential.

7.4 Error Analysis: Metric Type, Language, and Modality

SASB metrics broadly fall into two categories: *quantitative* requirements, which demand explicit numerical values, and *discussion and analysis* re-



Figure 1: An example page from a Chinese ESG report, illustrating discussion-oriented disclosure through visual structure. For readability, the figure uses an English-labeled version of the original page.

quirements, which involve qualitative explanations of operations, policies, or risk management practices. The relative difficulty of these two requirement types varies systematically across languages due to differences in how information is distributed across textual and visual modalities.

In English and French disclosures, quantitative metrics are more error-prone because numerical values, units, and scopes are often embedded across descriptive prose rather than explicit scalar tables. Conversely, discussion-oriented requirements in these languages use recognizable policy prose, making them easier for text-based models to identify. Chinese disclosures present the opposite pattern: quantitative metrics are typically presented in structured tables, whereas discussion and analysis requirements rely heavily on visual layouts—such as maps, charts, and aligned graphics—with minimal continuous text (Figure 1).[4]

Text-dominant models often fail to detect discussion-oriented SASB compliance when the evidence is conveyed through visual structure, leading to systematic false negatives in Chinese even for substantively complete disclosures. This stems less from language ambiguity than from a modality shift. These findings reveal a limitation of English-centric, text-only ESG evaluation: when discussion is assumed to appear as narrative prose, visually expressed compliance becomes invisible. Addressing this requires multimodal datasets and models that reflect how different reporting cultures distribute compliance evidence across text and visuals.

8 Conclusion

By aligning industry-specific SASB metrics with page-level evidence across six languages, and by explicitly annotating disclosure completeness, the proposed MM-COMPLIANCE enables fine-grained analysis beyond topic identification or promise detection. Through two complementary tasks, we evaluated current vision-language models under realistic auditing settings. The results show that while image-native retrieval can often localize relevant evidence pages within long ESG reports, models consistently struggle to verify partial compliance, frequently collapsing incomplete disclosures into negative decisions. Performance further varies across languages and industries, reflecting differences in disclosure structure, layout conventions, and domain-specific reporting practices. Overall, our findings highlight that the primary bottleneck in automated sustainability auditing lies not in locating candidate evidence, but in requirement-aware completeness verification. MM-COMPLIANCE provides a benchmark for studying this challenge and offers a foundation for future work on framework-aware, multilingual, and multimodal ESG auditing systems that support transparent compliance assessment and long-term commitment tracing.

[4] https://esg.tsmc.com/zh-Hant/file/public/2024-TSMC-Sustainability-Report-c.pdf

Limitations

This work has several limitations related to dataset scope, annotation granularity, sampling strategy, evaluation framework, and baseline coverage.

First, while MM-COMPLIANCE covers six languages, the dataset is not intended to be exhaustive in either linguistic or geographic coverage. The availability of publicly released ESG reports, differences in reporting practices, and industry composition across regions may introduce sampling biases. Expanding the dataset to additional languages and regulatory contexts would improve generalizability.

Although Thai and Korean subsets are smaller due to regional SASB adoption constraints (Section 3.4), two methodological safeguards ensure evaluation validity:

1. Task 1 pairs positive page with a strictly sampled nearby negative from the same report, effectively getting active evaluation instances and reducing sampling noise.

2. Empirical results (Sections 6.1 and 6.2) show that key trends—such as RAG-induced performance drops and near-zero partial compliance F1 scores—are consistently preserved across all six languages. This confirms that evaluations reflect systemic task complexity rather than sample variance.

Second, our annotations operate at the page level. Although page-level evidence is useful for practical auditing workflows, compliance verification often requires finer-grained grounding (e.g., the exact table cell, chart element, or paragraph span that satisfies a requirement). In particular, discussion-and-analysis evidence may be embedded in complex visual layouts, and quantitative values may depend on scope and unit definitions scattered across a page. Future work could extend MM-COMPLIANCE with region-level annotations and structured evidence extraction.

Third, the completeness label captures whether a disclosure satisfies a SASB metric at the semantic level, but it does not enforce a single universally measurable notion of sufficiency. Some metrics require judgement about whether disclosed details are adequate (e.g., scope, methodology, or narrative specificity), which may introduce ambiguity in borderline cases despite careful annotation guidelines.

Fourth, Task 1 relies strictly on nearby negative pages to evaluate localized compliance reasoning. While effective for injecting hard negatives, this design prioritizes local visual and textual signals over macro-level document variations and may not capture model behavior against distant, topically unrelated pages. Models optimized solely on proximal negatives could exhibit calibration shifts in unrestricted document pipelines. Future extensions will incorporate a balanced mix of proximal hard negatives and distant easy negatives to better map global retrieval dynamics.

Fifth, it is important to clarify that MM-COMPLIANCE evaluates the completeness and framework-conformance of sustainability disclosures rather than verifying actual corporate sustainability performance or substantive outcomes (e.g., net emission reductions). A fully standard-compliant disclosure does not automatically imply superior ESG performance. Furthermore, because our dataset maps specific language subsets to distinct jurisdictions (e.g., Chinese-language reports to Taiwan) , the observed cross-lingual variances reflect an intertwined mix of linguistic complexity, local regulatory environments, institutional adoption of SASB, and regional corporate document design cultures. Future work should decouple purely linguistic challenges from country-specific reporting frictions.

Finally, MM-COMPLIANCE focuses on SASB-aligned disclosure auditing. While SASB provides a strong investor-oriented framework, real-world sustainability reporting may follow multiple standards simultaneously, and companies may use localized terminology or formats that do not align cleanly with metric phrasing. Bridging this framework-to-practice gap remains an open challenge for automated multilingual ESG auditing.

Ethical Considerations

Our study builds the dataset exclusively from publicly available ESG reports published by corporations, and thus it contains no personally identifiable or sensitive human data; the corpus represents corporate disclosures intended for public communication, and all experiments evaluate models on these materials for research purposes only. Before the annotation process, we fully informed the annotators about the nature and goals of the task and obtained their informed consent. Annotation protocols varied by language team. For the English and French subsets, the data collection framework was approved by the Legal Department. For the

Thai subset, four native university students were recruited and compensated THB 200 per sample after receiving in-person instructions and annotated examples. The Korean subset was annotated by research interns at the financial engineering laboratory as part of their internship curriculum, with guidelines communicated directly within the team. The Japanese subset was annotated based on the discussion of an ESG financial expert and a financial department student. Formal IRB approval was not obtained for the Asian language subsets, as the source material consists entirely of public corporate disclosures.

During the preparation of this work, the authors used generative AI solely for grammar and style suggestions. After using this tool, the authors reviewed and edited the content, taking full responsibility for the publication's content.

Acknowledgment

The work was partially supported by the JSPS the Grant-in-Aid for Scientific Research (B) (#23K28375). The NTU-affiliated authors acknowledge support from NSTC Grants NSTC 114-2221-E-002-070-MY3 and NSTC 115-2634-F-002-012, and Grant 115L900901. Chung-Chi Chen's contribution to this work was supported by the NII Institute's Internal Open-Call Research Project, Grant No. 4070050026.

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A Detailed Annotation Protocol

To ensure the reproducibility and reliability of the MM-COMPLIANCE dataset, this appendix provides comprehensive details regarding our annotation guidelines, the qualifications of the annotating teams across different languages, and qualitative case studies illustrating the application of the protocol.

A.1 Annotation Guidelines

The design of the multilingual annotation guidelines was led by a professional Data & Language Analyst to maintain a harmonized taxonomy and consistent standards across all language teams. The framework operates by evaluating the compliance of corporate environmental, social, and governance (ESG) disclosures against the Sustainability Accounting Standards Board (SASB) criteria at a page-level semantic layer.

An ESG disclosure within a page is decomposed into a **Statement** (a declaration about a policy or action) and **Evidence** (concrete elements that substantiate the claim). To prevent vague or unsubstantiated claims, particular emphasis is placed on verifiable evidence, structured around three core target fields:

- **Value**: The primary evidence extracted from the text fragment or table. It represents the numerical target metric (excluding the year) or is labeled as *N/A* if the required information is purely qualitative.
- **Unit**: The unit of measurement associated with the value, such as percentages (%), presentation currencies, or specific physical quantities (e.g., metric tons, cubic meters).
- **Complete**: A boolean label evaluating whether the extracted *Value*, *Unit*, and associ-

ated scope or procedural details jointly fulfill the structural and metric-specific criteria expected under the *SASB Unit of Measurement* for that given sector.

A disclosure is labeled as **Complete** (`TRUE`) if the reported data directly mirrors or satisfies the standard definition in terms of quantitative values, operational scopes, or procedural details. If a metric is mentioned qualitatively without required numerical benchmarks, or if it lacks mandatory components, it is strictly categorized as **Incomplete** (`FALSE`).

Furthermore, uniform extraction challenges were governed by a unified protocol across teams: (1) if multiple digital values exist in a section, only the first relevant digital value corresponding to the SASB metric is evaluated; (2) when evaluating qualitative requirements (e.g., discussion and analysis metrics), `Complete` is set to `TRUE` as long as the topic's presence is verified, with *Value* and *Unit* recorded as *N/A*; (3) a value that matches units but possesses no thematic relevance to the specified SASB metric is explicitly marked as `FALSE`.

To ensure strict adherence to these criteria, all annotators followed a standardized, multi-stage workflow during the disclosure alignment process. The complete procedural lifecycle of this data annotation pipeline is illustrated in Figure 2.

To maintain cross-lingual consistency and eliminate subjective adjudication, ground-truth labels for the downstream Page-level Compliance Judgment task were programmatically derived from the segment-layer binary assessments. Disclosures fully satisfying SASB criteria (**Complete** = `TRUE`) were assigned *yes*, whereas those mentioning a metric but lacking required quantitative values or units (**Complete** = `FALSE`) were mapped to *yes but not complete*. Negative samples (*no*) were algorithmically constructed by taking each positive instance (both *yes* and *yes but not complete*), identifying the nearest non-disclosing page within the same report, and reassigning its page index. This rule-based mapping prevents annotator bias across language teams.

A.2 Annotator Qualifications and Quality Control

Annotations were carried out strictly by native speakers of the target languages or individuals with professional-level linguistic and operational proficiency in the respective languages. To capture diverse geopolitical and regional disclosure conventions, separate language teams were coordinated under a centralized protocol, with specific compositions as follows:

- **English and French Subsets**: Annotated by two professional Data and Language Analysts in their capacity within the laboratory framework.
- **Thai Subset**: Annotated by four native university students who received rigorous, in-person instructional training with annotated baseline examples, and were systematically compensated at a rate of THB 200 per sample.
- **Korean Subset**: Managed and annotated by specialized research interns at a financial engineering laboratory as a core component of their financial data curriculum.
- **Japanese Subset**: Executed via an iterative collaborative framework involving an ESG financial expert and a university student specializing in financial data tracking.
- **Chinese Subset**: Annotated by eight native Chinese-speaking research assistants within an information management and green financial technology laboratory framework, as part of their laboratory research activities.

To ensure robustness against the inherent subjectivity of evaluating compliance and checking subtle greenwashing, cross-lingual alignment sessions were established to standardize the semantic boundary of the *yes but not complete* label.

Furthermore, we conducted a rigorous Inter-Annotator Agreement (IAA) experiment. Because a single target SASB metric can frequently map onto multiple disparate fragments or tabular components across a visually dense layout, conventional single-label metrics like Cohen's Kappa are mathematically poorly suited. Instead, we applied the region-based **Jaccard coefficient** to evaluate the spatial overlap of the bounding regions and textual spans extracted manually by independent annotators. The manual baseline evaluation reached an aggregate spatial and semantic agreement score of **0.65** for Japanese and **0.61** for Korean, demonstrating strong consistency across native auditing teams. This variation reflects language-specific structural characteristics, where evidence in Korean ESG reports is sparsely distributed across multi-column tables, infographics, and fragmented visual elements,

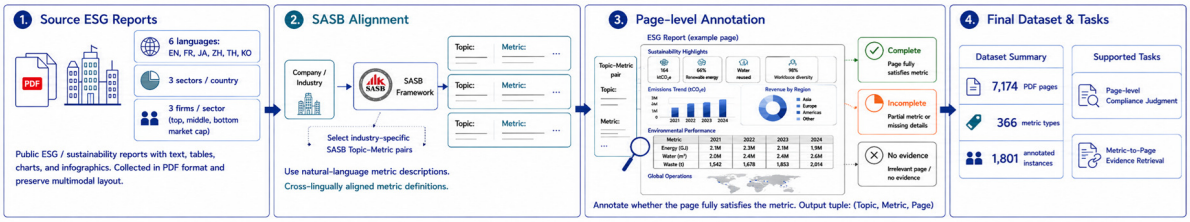


Figure 2: The structured workflow of the MM-COMPLIANCE annotation protocol.

making exact region-level alignment particularly challenging.

B Additional Evaluation across Open-Source and Proprietary VLMs

B.1 Analysis and Discussion on Task 1 (Page-level Compliance Judgment)

To assess architecture generalizability, we also evaluated two open-weight models (Qwen3-VL-8B-Instruct, Llama 3.2 Vision), and a proprietary state-of-the-art model (Gemini 3.5 Flash) in a zero-shot setting. The results mirror the structural challenges observed in our primary RAG experiments in Table 2 and Table 4 while highlighting a clear performance gap between open-weight and commercial model. The inclusion of the proprietary state-of-the-art model (Gemini 3.5 Flash) serves not as a deployment recommendation under privacy-sensitive conditions, but strictly as a frontier proprietary comparison model to quantify the remaining capability gap between open-weight architectures and frontier commercial models.

Commonalities with RAG-Augmented Baselines:

- **Partial Compliance Bottleneck**: All models severely struggle with partial disclosures (*yes but not complete*), yielding near-zero F1 scores (e.g., 0.030 for Qwen3-VL on image). This confirms that neither expanded context (via RAG) nor instruction tuning resolves the fundamental reliance on surface keyword matching over requirement reasoning.
- **Modality Sensitivity**: For table- and infographic-heavy layouts (e.g., Chinese and Korean subsets), direct image inputs consistently outperform OCR text across all evaluated models except Llama 3.2 vision.

Divergence Between Open-Weight and Proprietary Models:

- **Superiority in Fine-Grained Disambiguation**: Gemini 3.5 Flash achieves the highest overall macro-F1 (0.544 on Image) and retains a noticeably stronger class-wise F1 score of 0.230 on partial compliance, indicating better differential reasoning for missing required metrics.
- **Open-Weight Model Variance**: Qwen3-VL-8B-Instruct achieves high overall image accuracy (0.611) competitive with Gemini in binary classification (*yes* vs. *no*), yet collapses on partial disclosures. Conversely, Llama 3.2 Vision lags significantly (Image Acc: 0.424), constrained by complex visual layout parsing, while class-wise F1 score for partial compliance using OCR text retains 0.265.

B.2 Analysis and Discussion on Task 2 (Metric-to-Page Evidence Retrieval)

Methodology and Token Constraints. Evaluating generative VLMs on multi-page document retrieval requires managing context window constraints. Gemini 3.5 Flash processed full corporate reports natively within its 1M-token context window. For Qwen3-VL-8B-Instruct, context limits necessitated a two-stage chunking retrieval pipeline: partitioning documents into 80-page chunks ($\approx$ 90k tokens), extracting top-5 candidate pages per chunk, and re-ranking the merged candidates to select the final top-20 pages. Llama 3.2 Vision was excluded due to its 128k context limit being insufficient for full-report indexing.

Comparative Analysis and Common Findings. Evaluating Qwen3-VL-8B-Instruct (Table 10) and Gemini 3.5 Flash (Table 11) alongside the ColQwen2.5 baseline (Table 3) demonstrates retrieval generalizability across multi-page ESG disclosures. Across six languages, the results reveal distinct architectural trade-offs and shared operational behaviors across open-weight, proprietary, and late-interaction vision-language paradigms.

				Macro-averages			Yes			Yes but not complete			No		
ID	Format	Language	Acc.	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
1	text	English	0.622	0.554	0.505	0.487	0.581	0.693	0.632	0.400	0.065	0.111	0.681	0.756	0.717
2	text	French	0.615	0.643	0.479	0.470	0.590	0.888	0.709	0.667	0.087	0.154	0.673	0.463	0.548
3	text	Chinese	0.683	0.465	0.496	0.477	0.655	0.809	0.724	0	0	0	0.740	0.679	0.708
4	text	Thai	0.661	0.464	0.502	0.473	0.583	0.840	0.689	0	0	0	0.809	0.667	0.731
5	text	Japanese	0.508	0.392	0.471	0.392	0.361	0.838	0.504	0	0	0	0.814	0.574	0.673
6	text	Korean	0.490	0.408	0.433	0.384	0.379	0.490	0.427	0.250	0.026	0.048	0.595	0.783	0.676
	text	Avg.	0.597	0.488	0.481	0.447	0.525	0.760	0.614	0.220	0.030	0.052	0.719	0.654	0.676
7	image	English	0.597	0.403	0.471	0.434	0.556	0.682	0.612	0	0	0	0.652	0.732	0.690
8	image	French	0.635	0.777	0.485	0.466	0.591	0.910	0.717	1.000	0.044	0.083	0.741	0.500	0.597
9	image	Chinese	0.735	0.512	0.533	0.512	0.672	0.921	0.777	0	0	0	0.864	0.679	0.760
10	image	Thai	0.587	0.404	0.449	0.415	0.547	0.820	0.656	0	0	0	0.667	0.526	0.588
11	image	Japanese	0.485	0.389	0.458	0.375	0.352	0.865	0.500	0	0	0	0.816	0.508	0.626
12	image	Korean	0.624	0.662	0.566	0.510	0.494	0.863	0.629	0.667	0.053	0.098	0.825	0.783	0.803
	image	Avg.	0.611	0.525	0.494	0.452	0.535	0.844	0.649	0.278	0.016	0.030	0.761	0.621	0.677

Table 7: Task 1 evaluation results using Qwen3-VL-8B-Instruct across different modalities and languages.

				Macro-averages			Yes			Yes but not complete			No		
ID	Format	Language	Acc.	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
1	text	English	0.493	0.510	0.478	0.424	0.739	0.193	0.306	0.261	0.387	0.312	0.530	0.854	0.654
2	text	French	0.375	0.396	0.354	0.333	0.588	0.225	0.325	0.088	0.261	0.132	0.511	0.575	0.541
3	text	Chinese	0.524	0.480	0.453	0.438	0.640	0.360	0.460	0.077	0.250	0.118	0.724	0.750	0.737
4	text	Thai	0.504	0.469	0.460	0.435	0.640	0.320	0.427	0.179	0.357	0.238	0.588	0.702	0.640
5	text	Japanese	0.577	0.543	0.526	0.529	0.571	0.432	0.492	0.414	0.375	0.393	0.644	0.771	0.702
6	text	Korean	0.483	0.463	0.457	0.428	0.474	0.177	0.257	0.395	0.395	0.395	0.522	0.800	0.632
	text	Avg.	0.493	0.477	0.455	0.431	0.609	0.285	0.378	0.236	0.338	0.265	0.587	0.742	0.651
7	image	English	0.453	0.387	0.379	0.320	0.500	0.171	0.254	0.200	0.065	0.098	0.460	0.902	0.609
8	image	French	0.412	0.314	0.325	0.253	0.524	0.124	0.200	0	0	0	0.417	0.850	0.560
9	image	Chinese	0.455	0.384	0.356	0.292	0.588	0.112	0.189	0.100	0.063	0.077	0.463	0.893	0.610
10	image	Thai	0.463	0.423	0.348	0.278	0.600	0.060	0.109	0.200	0.071	0.105	0.469	0.912	0.619
11	image	Japanese	0.385	0.298	0.297	0.257	0.067	0.027	0.039	0.400	0.125	0.191	0.429	0.738	0.542
12	image	Korean	0.376	0.324	0.320	0.231	0.400	0.039	0.071	0.182	0.053	0.082	0.391	0.867	0.539
	image	Avg.	0.424	0.355	0.338	0.272	0.447	0.089	0.144	0.180	0.063	0.092	0.438	0.860	0.580

Table 8: Task 1 evaluation results using Llama 3.2 Vision across different modalities and languages.

				Macro-averages			Yes			Yes but not complete			No		
ID	Format	Language	Acc.	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1	Prec.	Rec.	F1
1	text	English	0.552	0.482	0.480	0.473	0.651	0.466	0.543	0.162	0.194	0.177	0.638	0.781	0.700
2	text	French	0.563	0.510	0.495	0.487	0.674	0.719	0.696	0.159	0.304	0.209	0.698	0.463	0.556
3	text	Chinese	0.735	0.573	0.570	0.570	0.779	0.832	0.804	0.111	0.125	0.118	0.829	0.750	0.788
4	text	Thai	0.650	0.486	0.496	0.477	0.600	0.857	0.706	0.000	0.000	0.000	0.857	0.632	0.727
5	text	Japanese	0.562	0.472	0.506	0.480	0.482	0.703	0.571	0.150	0.094	0.115	0.786	0.721	0.752
6	text	Korean	0.544	0.503	0.487	0.454	0.424	0.490	0.455	0.444	0.104	0.170	0.642	0.867	0.738
	text	Avg.	0.601	0.505	0.506	0.490	0.601	0.678	0.629	0.171	0.137	0.131	0.742	0.702	0.710
7	image	English	0.532	0.490	0.479	0.466	0.686	0.398	0.504	0.186	0.258	0.216	0.598	0.781	0.677
8	image	French	0.573	0.536	0.527	0.511	0.718	0.685	0.701	0.177	0.391	0.243	0.714	0.500	0.588
9	image	Chinese	0.751	0.601	0.598	0.598	0.807	0.798	0.802	0.136	0.188	0.158	0.861	0.810	0.834
10	image	Thai	0.636	0.551	0.554	0.536	0.735	0.720	0.727	0.125	0.214	0.158	0.792	0.667	0.724
11	image	Japanese	0.639	0.582	0.585	0.579	0.511	0.622	0.561	0.400	0.313	0.351	0.833	0.820	0.826
12	image	Korean	0.631	0.574	0.583	0.572	0.554	0.706	0.621	0.320	0.211	0.254	0.848	0.833	0.840
	image	Avg.	0.627	0.556	0.554	0.544	0.668	0.655	0.653	0.224	0.262	0.230	0.774	0.735	0.748

Table 9: Task 1 evaluation results using Gemini 3.5 Flash across different modalities and languages.

ID	Format	Language	F1@1	F1@5	F1@20	Hit@5	Hit@20
7	image	English	0.353	0.264	0.129	0.721	0.849
8	image	French	0.310	0.310	0.183	0.795	0.945
9	image	Chinese	0.356	0.203	0.089	0.596	0.766
10	image	Thai	0.248	0.121	0.055	0.364	0.579
11	image	Japanese	0.275	0.191	0.086	0.575	0.713
12	image	Korean	0.227	0.244	0.115	0.635	0.778
	image	Average	0.295	0.222	0.110	0.614	0.772

Table 10: Task 2 overall retrieval performance using Qwen3-VL-8B-Instruct.

ID	Format	Language	F1@1	F1@5	F1@20	Hit@5	Hit@20
7	image	English	0.391	0.318	0.138	0.744	0.837
8	image	French	0.462	0.370	0.181	0.877	0.932
9	image	Chinese	0.649	0.375	0.133	0.989	1.000
10	image	Thai	0.546	0.259	0.080	0.777	0.835
11	image	Japanese	0.610	0.341	0.118	0.913	0.925
12	image	Korean	0.577	0.571	0.221	1.000	1.000
	image	Average	0.539	0.372	0.145	0.883	0.921

Table 11: Task 2 overall retrieval performance using Gemini 3.5 Flash.

- **Commonalities in Retrieval Dynamics:** All evaluated architectures display a consistent precision-recall trade-off as rank cutoff K expands: $F1@K$ contracts rapidly from $K = 1$ to $K = 20$, whereas $Hit@K$ increases substantially. Expanding from $Hit@5$ to $Hit@20$, coverage improves for ColQwen2.5 (0.719 to 0.882), Qwen3-VL-8B (0.614 to

0.772), and Gemini 3.5 Flash (0.883 to 0.921). Conversely, $F1@20$ drops sharply across all systems (ColQwen2.5: 0.139, Qwen3-VL: 0.110, Gemini: 0.145). These shared dynamics indicate that current VLMs function effectively as coarse candidate shortlisting mechanisms—isolating evidence within top-20 boundaries—but struggle to pinpoint the single authoritative page at rank 1.

- **Impact of Multi-Page Context Processing Strategies:** Global context integration provides significant retrieval advantages over fragmented multi-stage pipelines. Native full-document processing within a 1M context window enables Gemini 3.5 Flash to achieve superior early localization ($F1@1 = 0.539$), outperforming ColQwen2.5's late-interaction embedding ($F1@1 = 0.319$). Conversely, Qwen3-VL-8B-Instruct exhibits lower precision ($F1@1 = 0.295$, $Hit@5 = 0.614$), degraded by 80-page chunking boundaries and re-ranking heuristics. This confirms that context window continuity and image-native token alignment are essential to prevent information loss across multi-page ESG disclosures.